

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA104/MCA114: Advanced Database Management System (New/Old)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Compare and contrast lock-based and timestamp-based concurrences control. Check whether the schedule S is conflict serializable or not? 6+6
S: R3(y); R3(z); R1(x); W1(x); W3(y); W3(z); R2(z); R1(y); W1(y); R2(y); W2(y); R2(x); W2(x)
2. How can active databases be used in real-time applications like fraud detection or monitoring systems? What are the challenges and solutions for managing geospatial data in databases? 7+5
- 3(a) What are the different types of join algorithms used in query execution? Explain. 6
- (b) Explain the difference between cost-based and rule-based query optimizer. 6

Group B

Answer SIX questions.

6×6=36

4. Consider the following table:
EMPLOYEES (Employee_ID, First_Name, Last_name, Email, Phone_Number, Start_Date, Salary, Manager_Id, Department)
 - (a) Find the employee with the highest salary.
 - (b) Write a query to group employees by department and find the total no. of employees in each department.
 - (c) Display the details of employees who do not belong to "HR" department.
5. Explain the concept of parallel and distributed databases.
6. What is database security? Explain the challenges of database security.
7. What are the possible reasons of disk failures? How can data be recovered from disk crashes?
8. Explain the tier architecture of DBMS.
9. What is SOAP? Explain X-query and X-path with examples.
10. What is indexing? Briefly explain the types of indexes.
11. What do you mean by undo and redo logging? Define check point in the context of system recovery.